

- 1 List and Explain the Concept of general electrical energy conservation opportunities (ECOs).

Electrical Energy Conservation opportunities (ECO's) are strategies or methods used to reduce electricity consumption without compromising output or comfort. These can be implemented in industries, buildings or institutions.

Common ECO's:

(i) Lighting Systems:

- * use LED lamps instead of incandescent / CFL bulbs.
- * Implement motion sensors, dimmers, and timers
- * utilize natural daylight wherever possible.

(ii) Motor Efficiency:

- * Replace old motors with energy-efficient ones (IE3 or IE4)
- * use variable frequency drives (VFDs) to control motor speed.

(iii) Power Factor Correction:

- * Low power factor causes more current and energy loss.
- * use capacitor banks to correct the power factor

(iv) Efficient Transformers:

- * use low-loss transformers and operate them near their rated capacity to minimize iron and copper losses.

(v) HVAC Optimization:

- * Maintain proper insulation and use efficient chillers
- * operate during cooler hours and use thermostats and timers

(vi) preventive Maintenance:

- * Clean filters, fix air leaks, and lubricate moving parts regularly
- * prevents system inefficiencies due to wear and tear

a) Analyze preliminary, Targeted and Detailed Energy Audit

An Energy Audit is a Systematic study to identify where and how energy is used in a facility and to recommend cost-effective ways to reduce energy consumption.

(i) Preliminary Energy Audit:

* Objective: Quick diagnosis of major energy saving areas.

* Tools: Basic instruments or visual inspection.

* Output: List of low-cost/no-cost recommendations

* Example: checking if lights or fans are on unnecessarily.

(ii) Targeted Energy Audit:

* Objective: Focuses on a specific system (e.g., lighting, compressed air).

* Tools: portable energy meters, lux meters, thermometers.

* Output : Medium-level recommendations specific to that system

* Example : Study of compressor air leak losses.

(ii) Detailed Energy Audit :

* Objective : In-depth analysis of energy flow and performance

* Tools : Long-term data loggers, software tools (e.g., RET Screen, eQUEST)

* Output : Investment-grade recommendations with payback analysis.

* Example : Full audit of a plant's electrical, thermal, & mechanical systems.

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An industry has a daily requirement of 200 tons of Coal to meet its electrical energy and thermal energy requirements of 1.2×10^6 MJ and 1.6×10^6 MJ respectively. It uses a Cogeneration plant for this purpose. What is the overall efficiency of the plant? If it makes use of two separate plants for electrical and thermal power production instead of Cogeneration plant, what will be the combined overall efficiency? The efficiency of individual electrical and thermal plant is 30% and 80% respectively. The heating value of the coal may be assumed as 20 MJ/kg

Given :

- * daily coal use = 200 tons/day
- * calorific value of coal = 20 MJ/kg
- * Total energy from coal = $200 \times 1000 \times 20 = 4,000,000$ MJ/day
- * Electrical requirement = 1.2×10^6 MJ/day
- * Thermal requirement = 1.6×10^6 MJ/day

Cogeneration plant Efficiency:

It provides both electrical and thermal energy from the same fuel source.

$$\text{Total useful output} = 1.2 \times 10^6 + 1.6 \times 10^6 = 2.8 \times 10^6 \text{ MJ}$$

$$\text{Efficiency} = \frac{\text{Useful Energy output}}{\text{Energy input}} \times 100$$

$$= \frac{2.8 \times 10^6}{4.0 \times 10^6} \times 100$$

$$= 70\%$$

Separate plants Efficiency:

Given

$$\ast \text{ Electrical plant efficiency} = 30\%$$

$$\ast \text{ Thermal plant efficiency} = 80\%$$

$$\text{Energy required for electrical} = \frac{1.2 \times 10^6}{0.3} = 4.0 \times 10^6 \text{ MJ}$$

Energy required for thermal =

$$\frac{1.6 \times 10^6}{0.8} = 2.0 \times 10^6 \text{ MJ}$$

electrical + thermal energy:

$$4.0 \times 10^6 + 2.0 \times 10^6 =$$

Total energy $6.0 \times 10^6 \text{ MJ}$
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Overall efficiency (separate plants) =

$$\frac{2.8 \times 10^6}{6.0 \times 10^6} \times 100 = 46.67\%$$

Cogeneration plant efficiency (70%) is significantly higher than using separate systems (46.67%).

4) "Binary Cycle plant" is More Efficient than open cycle plant. Substantiate the statement.

A Binary Cycle power plant uses two working fluids. It operates by transferring heat from a primary heat source to a secondary (organic) fluid with a lower boiling point. This vaporizes and drives a turbine.

Why Binary Cycle is More Efficient:

(i) Heat Utilization:

- * It uses low-temperature sources like geothermal heat @ waste heat effectively.
- * In open cycle, low-grade heat is wasted.

(ii) Closed Loop:

- * The Secondary fluid is continuously recirculated in a closed loop $\rightarrow$ no loss of fluid $\rightarrow$ better heat retention

(iii) Flexibility:

- * Can work with different fluid combinations (like water + isobutane).
- * Suits areas where high-temperature sources aren't available.

(iv) Environmental Benefit:

- * No greenhouse gases emitted.
- * Ideal for renewable energy applications (geothermal)

(v) Overall Efficiency:

- * Combined cycle efficiencies $> 50\%$.
- * Open cycle thermal plants typically operate at 30-35% efficiency due to exhaust heat losses.